

Can vibrotactile stimulation drive brain clearance?

Wearable vibrotactile stimulation, brain rhythms, and glymphatic clearance
— rationale, plan, and first electrophysiology (in the mouse)

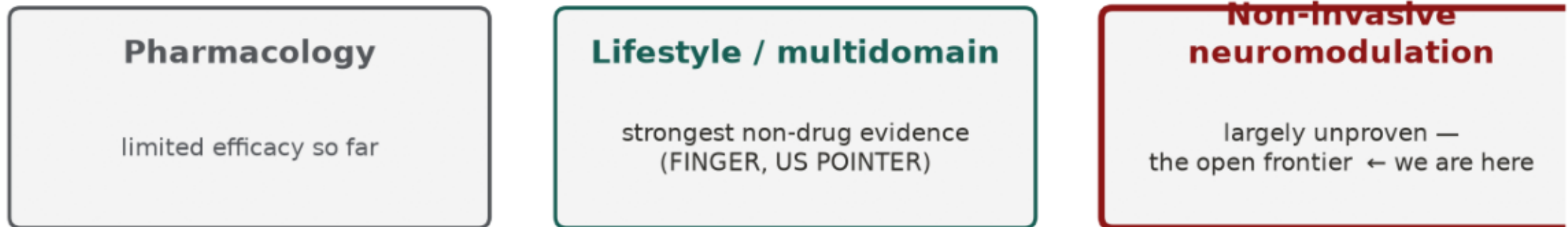
Pardis Miri · a Snyder Lab (Stanford) × Buzsáki Lab (NYU) collaboration



Part 1 · The vision — clearing the brain with vibrotactile stimulation

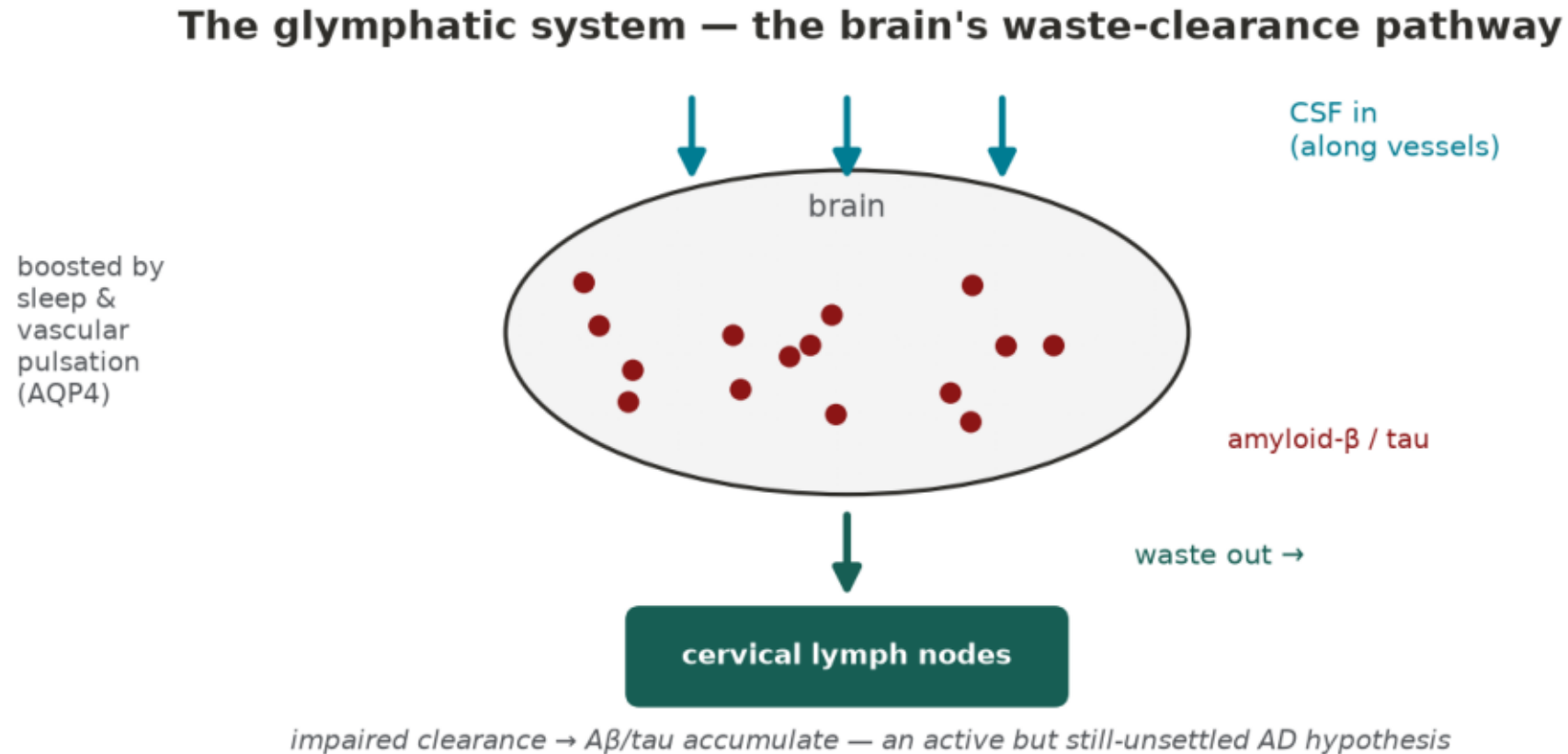
Alzheimer's resists drugs — non-invasive neuromodulation is an open frontier

Approaches to Alzheimer's — and the open frontier



The strongest non-drug clinical evidence today is multidomain lifestyle (FINGER, Ngandu 2015; US POINTER, Baker 2025; synthesis: Lancet Commission 2024). Single domains alone underperform. Non-invasive stimulation is largely unproven — which is the opportunity.

A leading hypothesis: failing glymphatic clearance lets amyloid build up



The brain clears amyloid- β / tau via CSF-interstitial exchange (the glymphatic system), boosted by sleep and vascular pulsation. Impaired clearance is a leading — but still-unsettled — explanation for AD (Iliff 2012; Xie 2013; DTI-ALPS, ADNI 2024).

And clearance can be driven non-invasively — 40 Hz in mice, focused ultrasound at Stanford

Clearance can be driven non-invasively — in animals



open question: can a WEARABLE modality do it?

40 Hz sensory stimulation drove ~4x CSF influx and ~30% less cortical amyloid, abolished when AQP4 is blocked (Murdock 2024, Nature). Focused ultrasound enhances CSF influx and clearance (Aryal 2022; Azadian 2025). The open question: can a wearable modality do the same?

The proposal: use vibrotactile stimulation to engage brain rhythms and drive clearance

The proposal — a mechanism hypothesis to test



a mechanism hypothesis — to be tested, not yet a treatment claim

Wearable vibrotactile stimulation → entrain brain rhythms → engage glymphatic clearance of amyloid-β / tau. A mechanism hypothesis — explicitly not yet a treatment claim.

Why vibrotactile, and why wearable: the white space next to light and sound

Vibrotactile is the open frontier — next to light & sound

Audiovisual 40 Hz (GENUS)

developed, evidence base —
but lab-bound, hard to wear
daily, not sleep-friendly

Vibrotactile (this program)

wearable · sleep-compatible
· pairs with sensing

← the white space

40 Hz light/sound (GENUS) is the developed approach but hard to wear daily. Vibrotactile stimulation is the open frontier — wearable, sleep-compatible, and pairs naturally with sensing.

Part 2 · What the literature says — and what to test

40 Hz sensory stimulation lowers amyloid and drives clearance — animals, and early humans

**GENUS / Tsai 2016: 40 Hz light/sound lowers amyloid. Murdock 2024 (Nature): 40 Hz gamma drives glymphatic A β clearance
Human gamma-sensory is safe with target engagement (Chan 2022).**

But a steady-state response is not the same as entraining the brain's own rhythm

A driven 40 Hz response $\neq$ engaging native gamma (Buzsáki/Soula; some studies find 40 Hz light fails to entrain native gamma in AD-model mice). We separate target engagement from disease modification.

Clearance is actively driven by neural, vascular & sleep dynamics — so 'brain-state' stimulation is plausible

CSF flow tracks NREM slow waves (Fultz 2019) and EEG delta (Hablitz 2019); neuronal dynamics actively direct CSF perfusion (Jiang-Xie-Kipnis 2024, Nature); human sleep clearance moves plasma A β /tau (Dagum 2026).

The vibrotactile anchor: 40 Hz vibration helps in mouse AD models, and cortex codes vibration frequency

40 Hz whole-body vibration reduces pathology and improves function in AD-model mice (Suk 2023). Somatosensory cortex faithfully encodes vibration frequency (Romo, Mountcastle) — vibrotactile input can carry a frequency code.

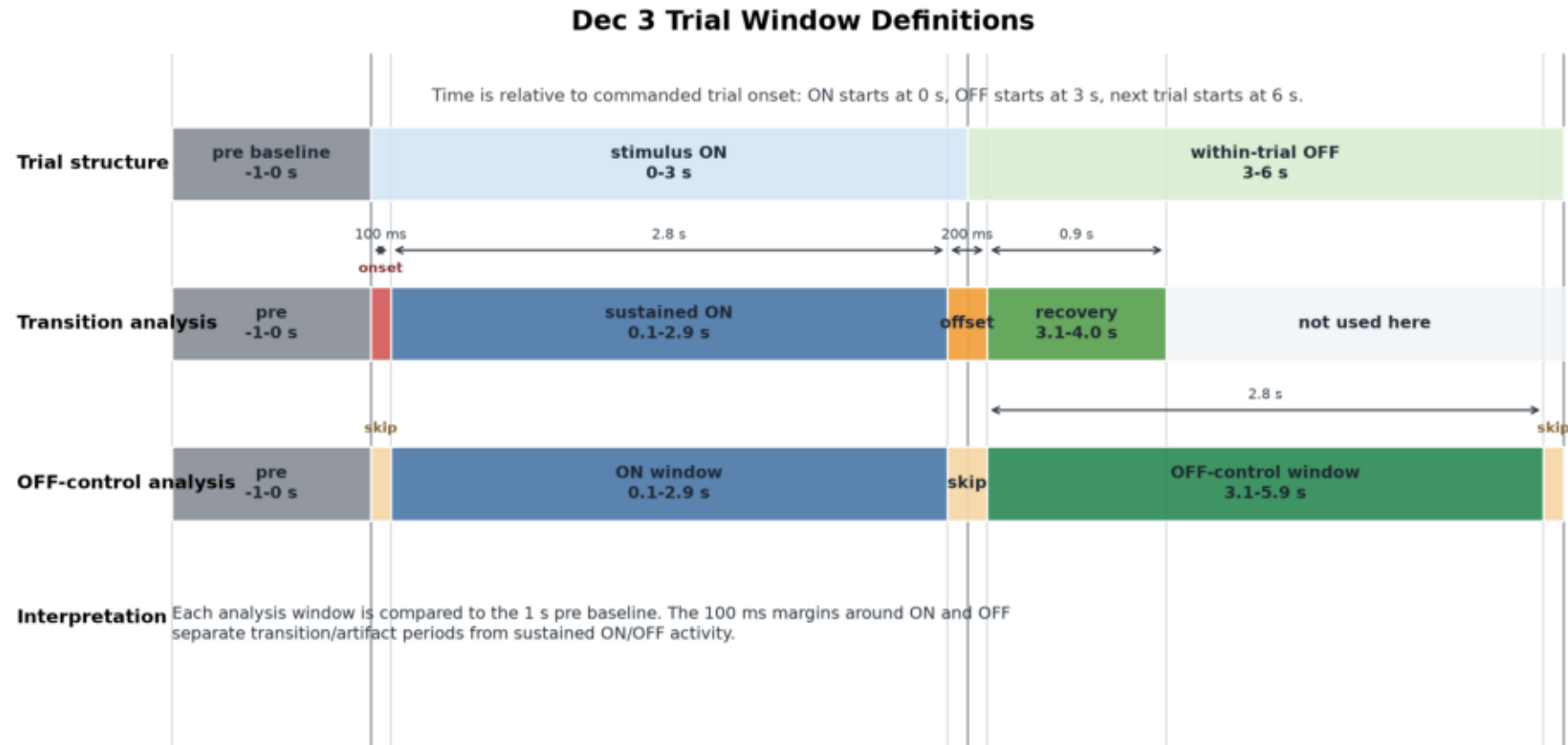
So which vibrotactile parameters should we test?

40 Hz (gamma) · ~50 Hz (our electrophysiology, Part 3) · slow ~1 Hz / sleep-band (clearance biology) · steady vs coordinated-reset patterning · controls: 20 Hz, sham, matched salience.

Part 3 · Does vibrotactile stimulation reach the brain? A first study

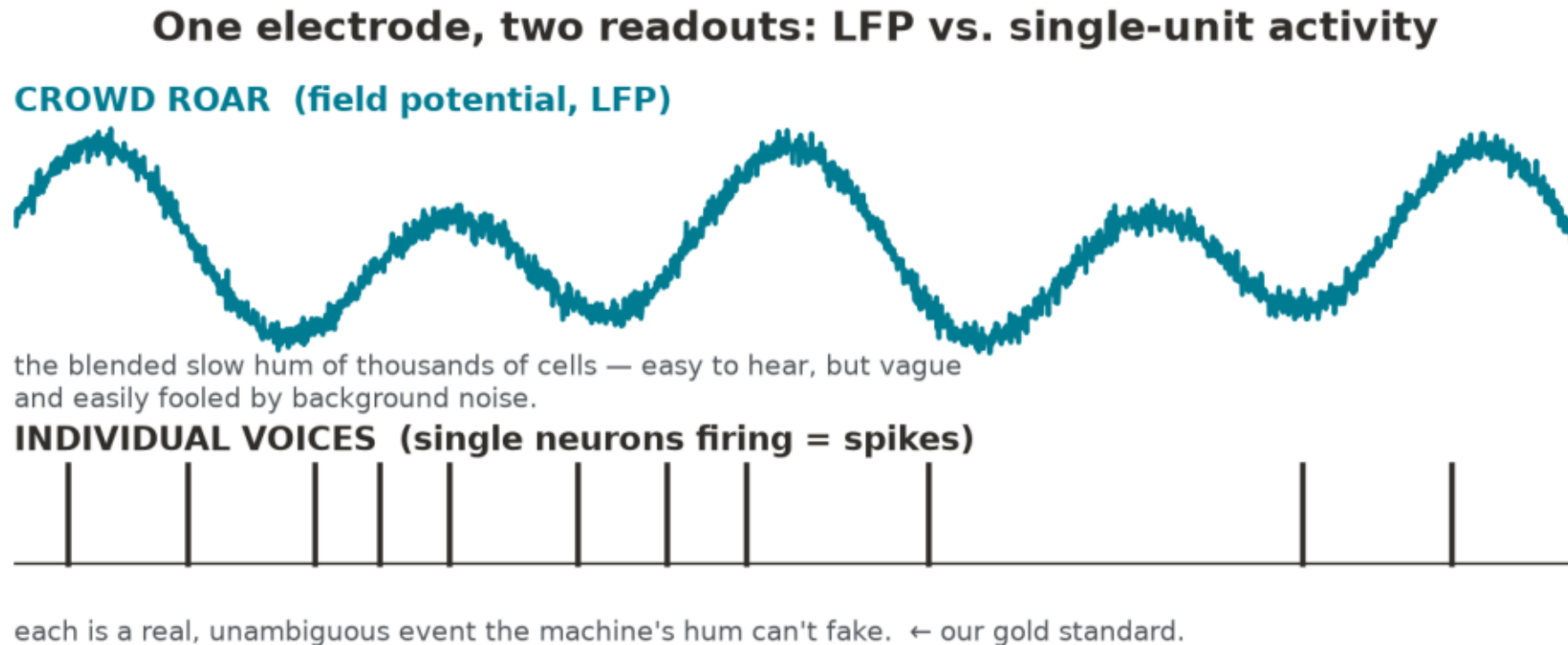
Foundational electrophysiology in the mouse — before any clearance claim

Design: parametric vibrotactile stimulation with simultaneous electrophysiology



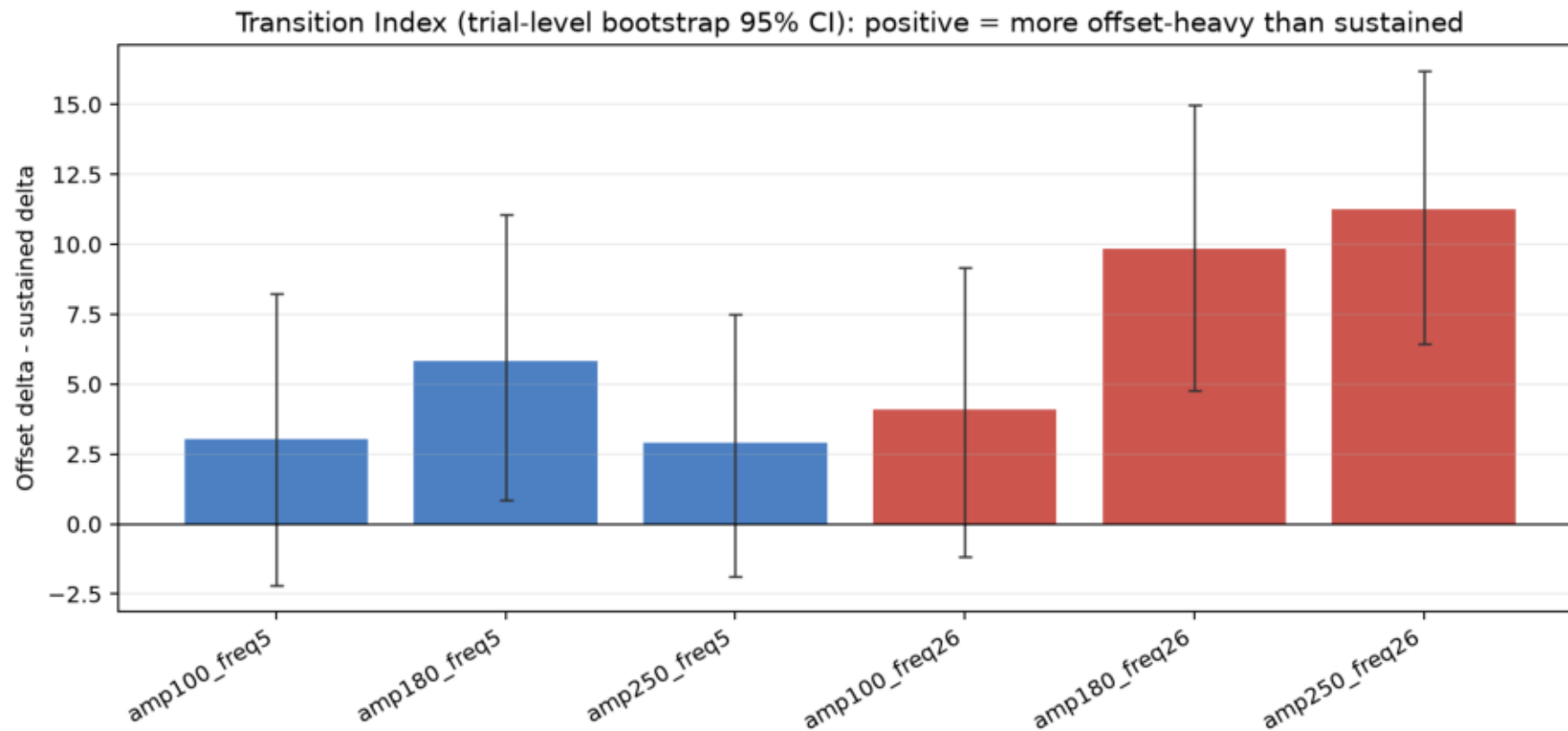
3 s ON / 3 s OFF blocks across four carrier frequencies (5/10/26/50 Hz) and three amplitudes; ~200 repeats/condition; dual-region linear-probe recording (hippocampus + entorhinal cortex).

One electrode yields two readouts: LFP and single-unit activity



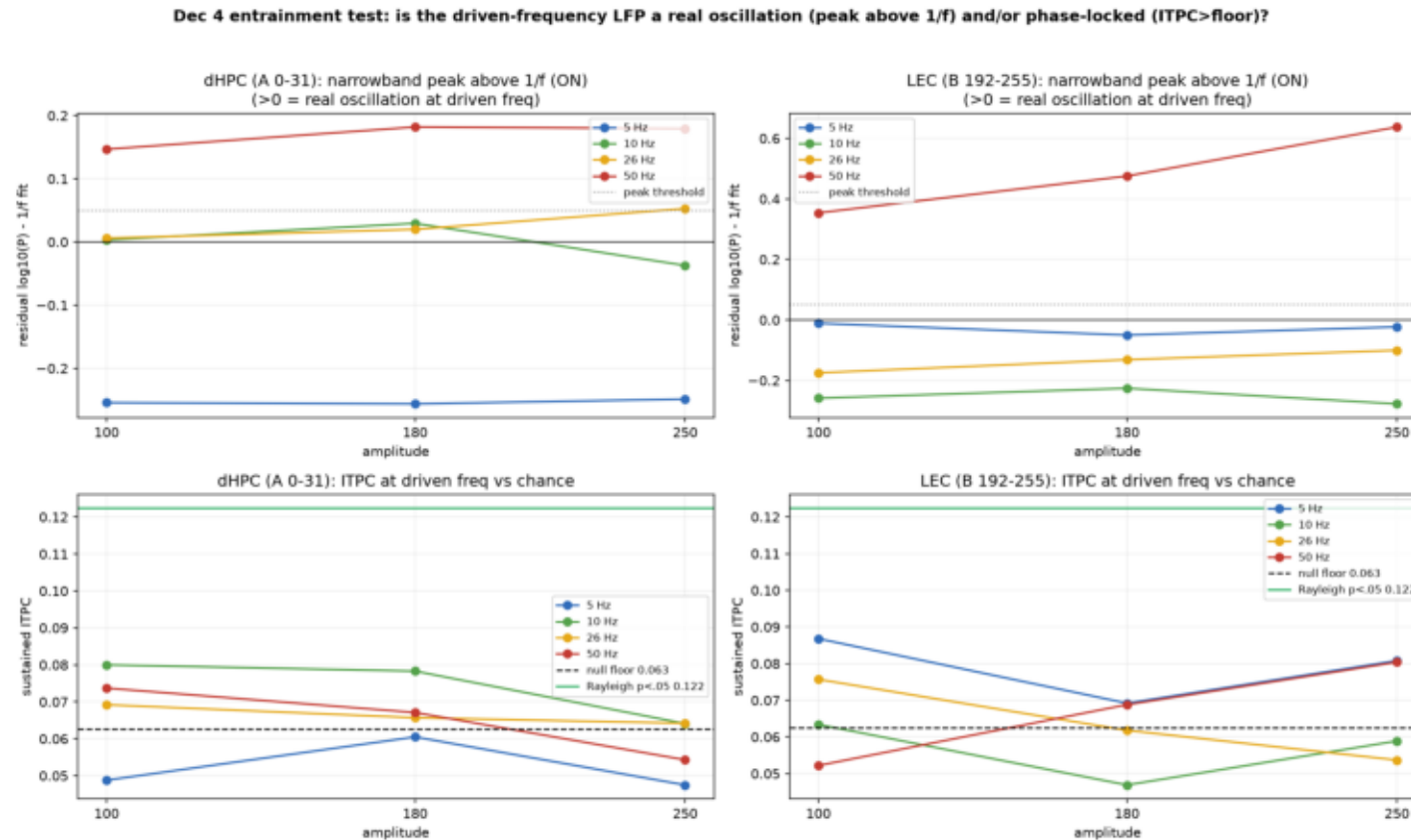
LFP = low-frequency aggregate field (many cells). A single unit = one individual neuron's firing (spikes), isolated by spike sorting. They differ in what can fake them — spikes are the conservative readout.

A robust LFP response — transition-weighted, not a sustained oscillation



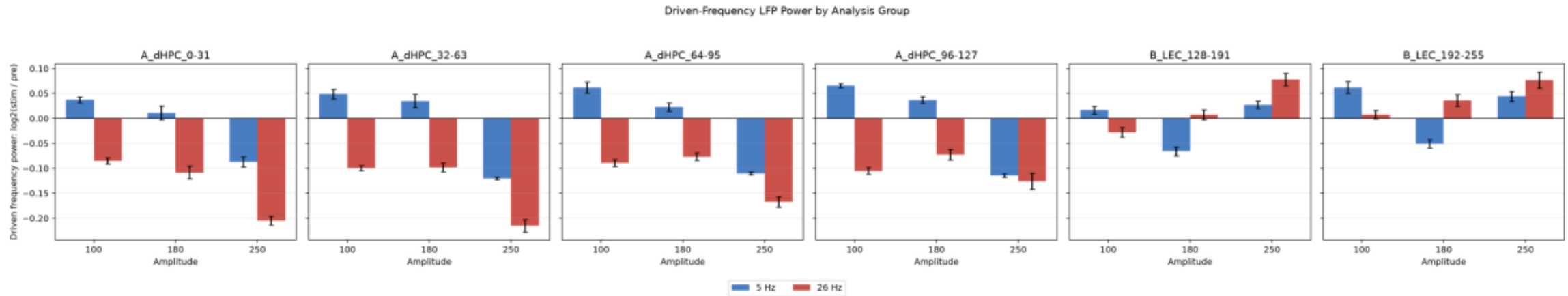
The broadband LFP responds at stimulus onset/offset (an evoked transient), not as a sustained rhythm at the carrier frequency.

No frequency-following in hippocampus at any carrier



No narrowband spectral peak above the $1/f$ background, and inter-trial phase consistency at chance — across 5/10/26/50 Hz, replicated across sessions.

Entorhinal cortex shows an amplitude-graded 50 Hz LFP increase



A narrowband 50 Hz LFP power increase, scaling with amplitude — the candidate entrainment signal. But the LFP is exactly the readout most vulnerable to artifact.

The confound: stimulator-coupled 50 Hz electrical artifact

The confound: stimulator-coupled 50 Hz electrical artifact

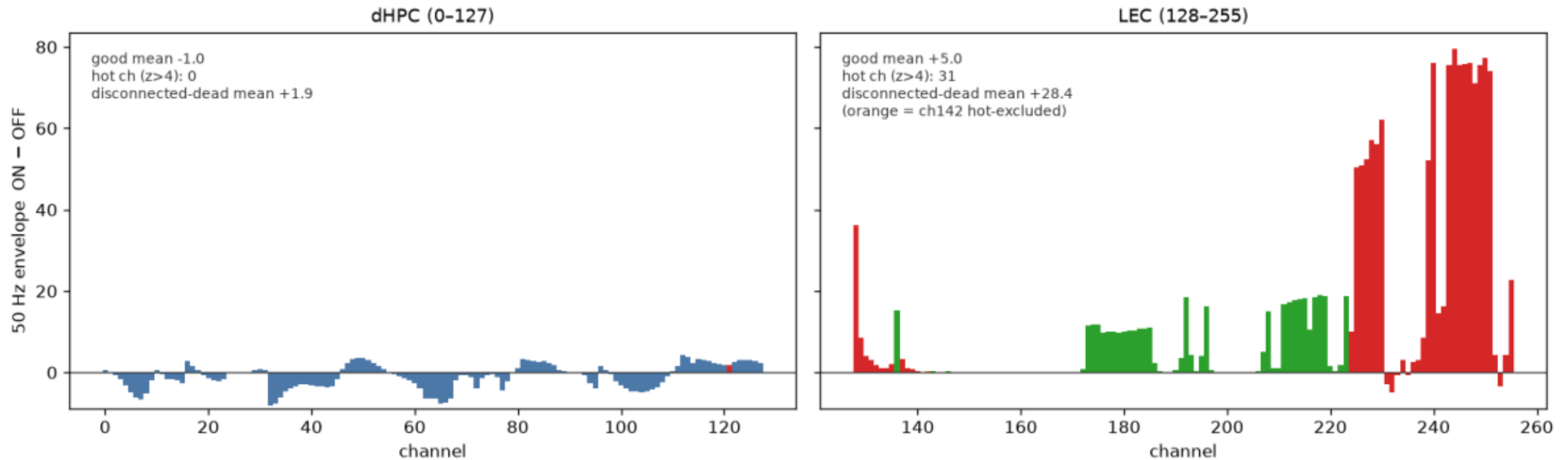


The mic records BOTH → a clean '50 Hz' could be the SPEAKER, not the brain.

An electromechanical actuator can inject a 50 Hz electrical artifact directly into the recording — indistinguishable from a neural 50 Hz in the LFP alone.

Disconnected channels carry the 50 Hz — so it is largely pickup

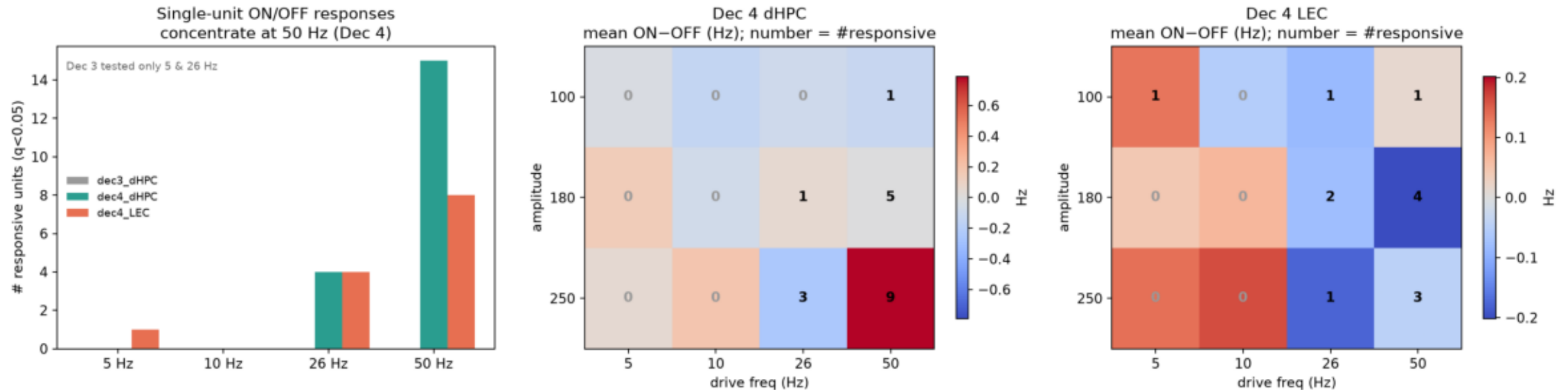
Is the 50 Hz pickup gradient LEC-only? dHPC (left): flat, ~ 0 , no hot channels, dead ch clean. LEC (right): rises toward the dead blocks (128-135, 224-255).



Disconnected electrodes (which cannot record neural signal) show $\sim 6\times$ more 50 Hz than in-tissue channels. The entorhinal 50 Hz LFP is substantially artifact.

Frequency-specific single-unit rate modulation at 50 Hz

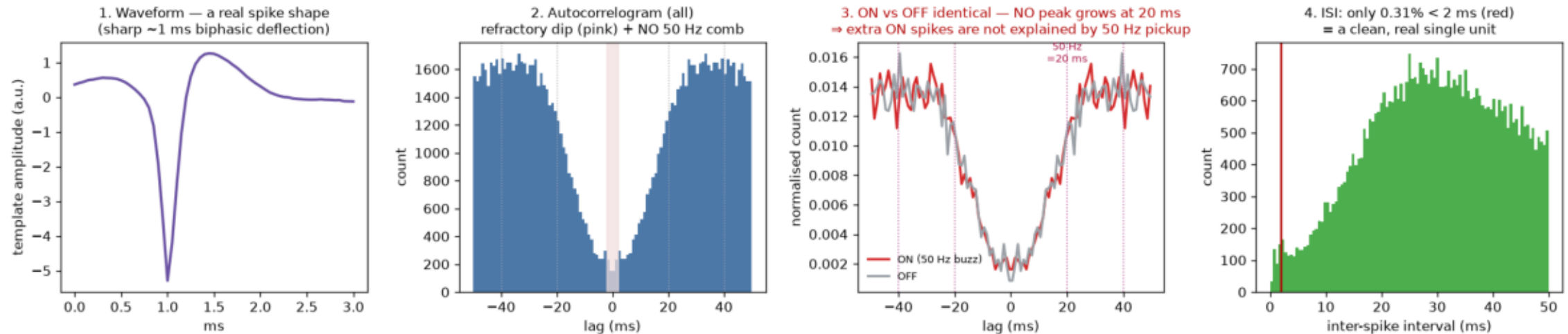
Curated single-unit ON-vs-OFF firing — Dec 3 null at 5/26 Hz; Dec 4 effect at 50 Hz / high amplitude



Sorted single units change firing rate during 50 Hz / high-amplitude stimulation, in both regions — an effect electrical pickup cannot produce.

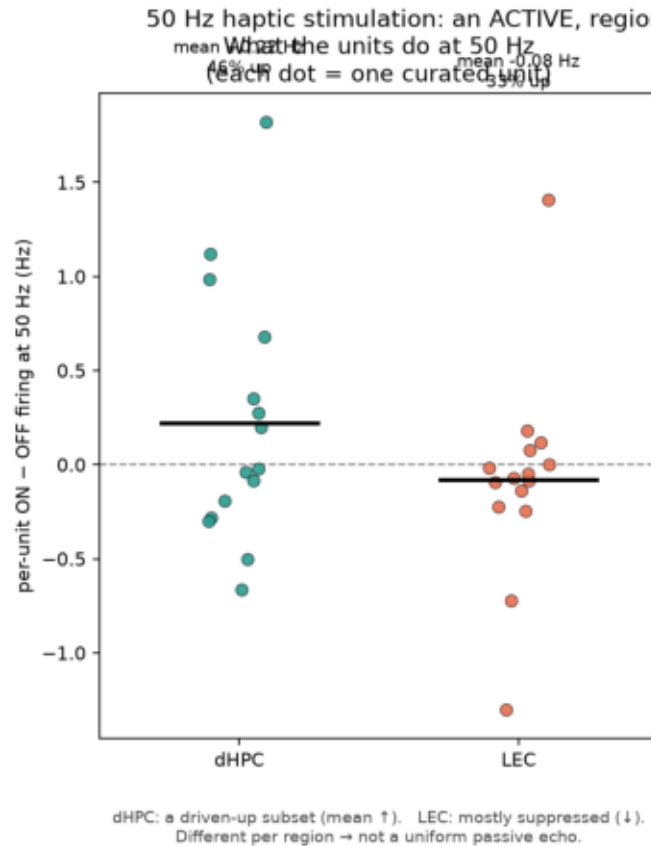
Controls: high-pass separation + autocorrelogram / ISI screens

UNIT 87 — the 'Phy view', computed: the soft-spot LEC unit is a REAL neuron, and its 50 Hz ON rate increase is not explained by 50 Hz pickup (its autocorrelogram grows no 50 Hz periodicity during ON).
Comprehensive screen: 0/8 up-going units show an ON 50 Hz comb, and 0/8 show an ON ISI<2ms rise $\Rightarrow$ the up-going rate increases are not explained by 50 Hz pickup.



Spikes are detected $> \sim 300$ Hz (50 Hz pickup is removed pre-detection); units show no stimulus-locked 50 Hz periodicity and no ON-rise in refractory violations.

Region-specific, bidirectional modulation — not a passive relay



What does the 50 Hz response mean? — the inference ladder

X Is it electrical / mechanical ARTIFACT?

NO — sorted single units change their firing RATE; pickup can't do that.

X Is it a PASSIVE ECHO (brain as a relay)?

UNLIKELY — regions respond DIFFERENTLY (dHPC driven-up subset; LEC suppressed) and the LFP 50 Hz is induced, not phase-locked. A relay would look the same everywhere.

✓ ACTIVE, region-specific processing?

SUPPORTED — the 50 Hz input is transformed differently by each circuit (not relayed).

? Are the regions COORDINATING ("working together")?

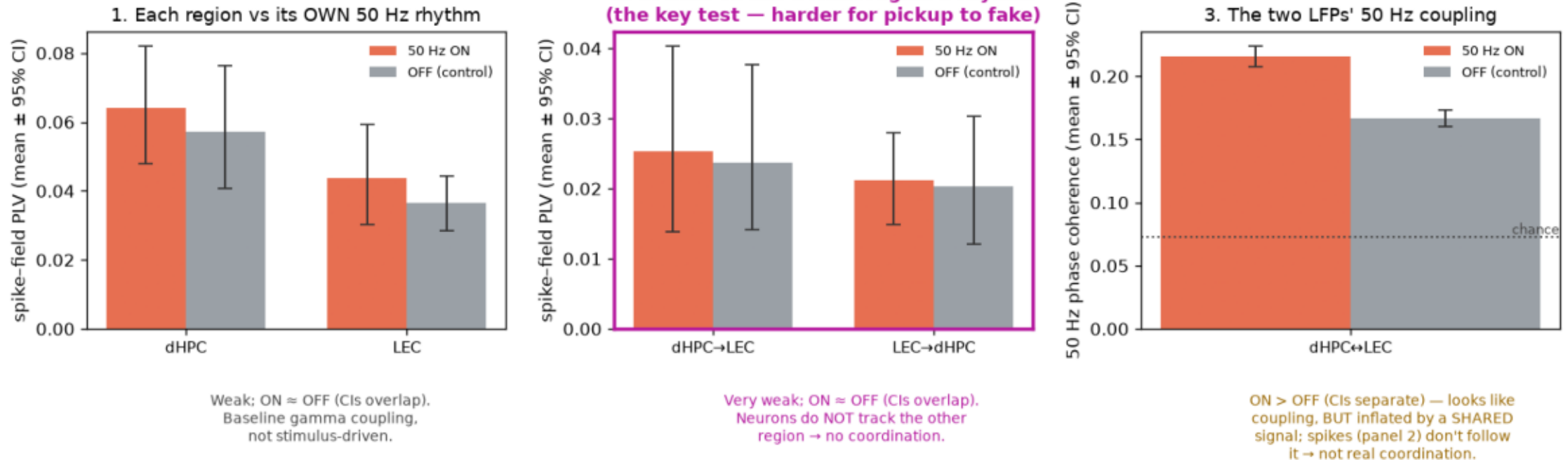
NOT TESTED — needs spike-field coupling + cross-region coherence. We showed each region responds, not that they coordinate.

Hippocampus: a driven-up subset. Entorhinal cortex: net-suppressed. Opposite transformations of identical input argue for active, circuit-specific processing.

No clear cross-regional coordination at 50 Hz

Do dHPC & LEC 'work together' at 50 Hz? — If they did, panel 2 (cross-region SPIKES) would rise during ON; it does NOT. Panel 3 (LFP) rises but is fooled by a shared signal. $\Rightarrow$ NO clear coordination.

2. SPIKES vs the OTHER region's rhythm (the key test — harder for pickup to fake)



50 Hz trials, all amplitudes pooled (amp100+180+250, 600 trials). ON = 3 s buzz; OFF = following 3 s gap.

LFP-LFP coherence rises, but the artifact-resistant cross-region spike-field measure does not — parsimoniously a shared signal, not interaction.

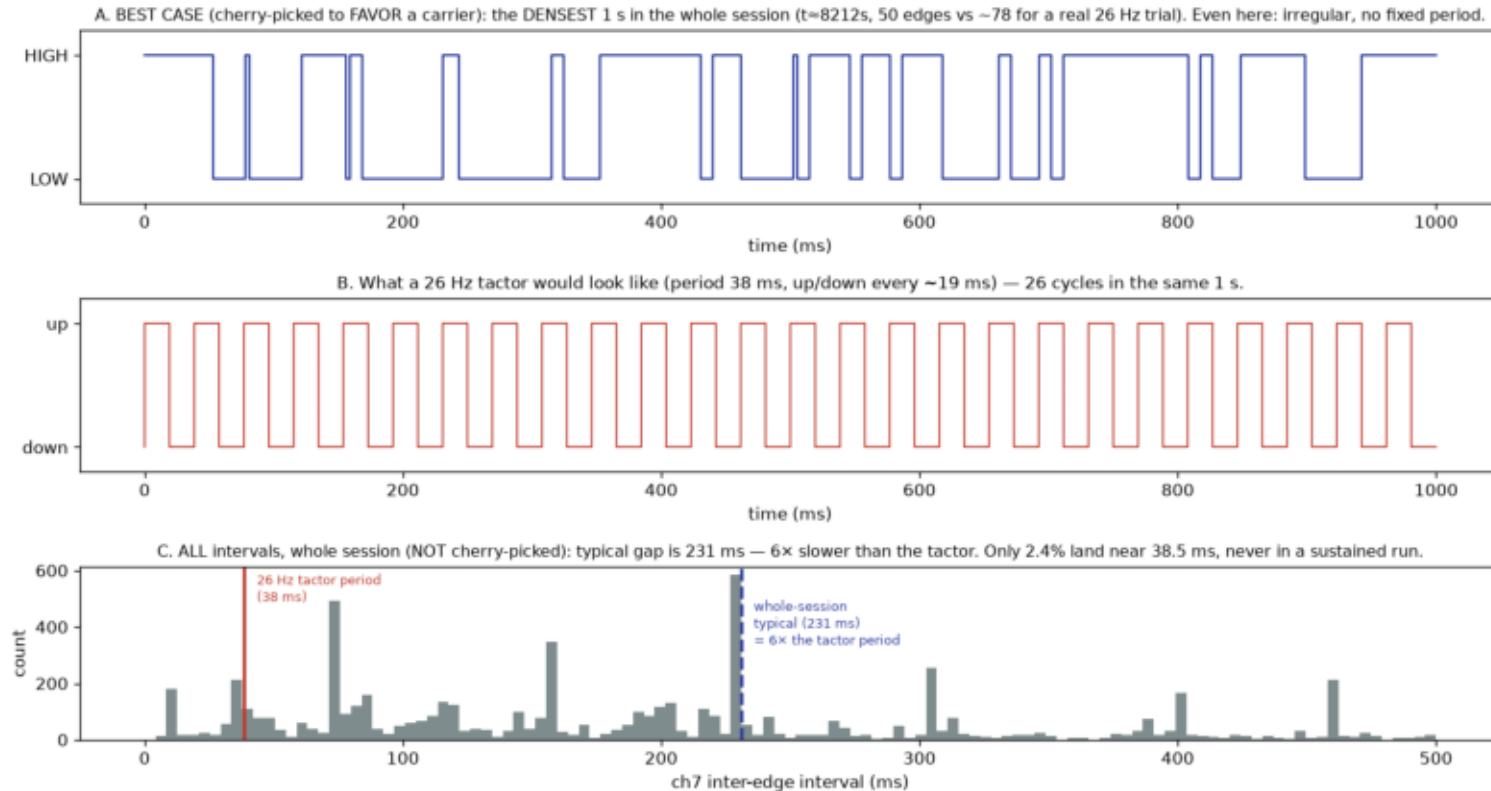
Part 4 · Limits, and the next study

Entrainment was untestable — a measurement gap, not a null

Our acquisition was not equipped to test entrainment: no time-aligned stimulus-phase reference was recorded — so phase-following is untestable here, not negative.

The digital sync channel never captured the carrier

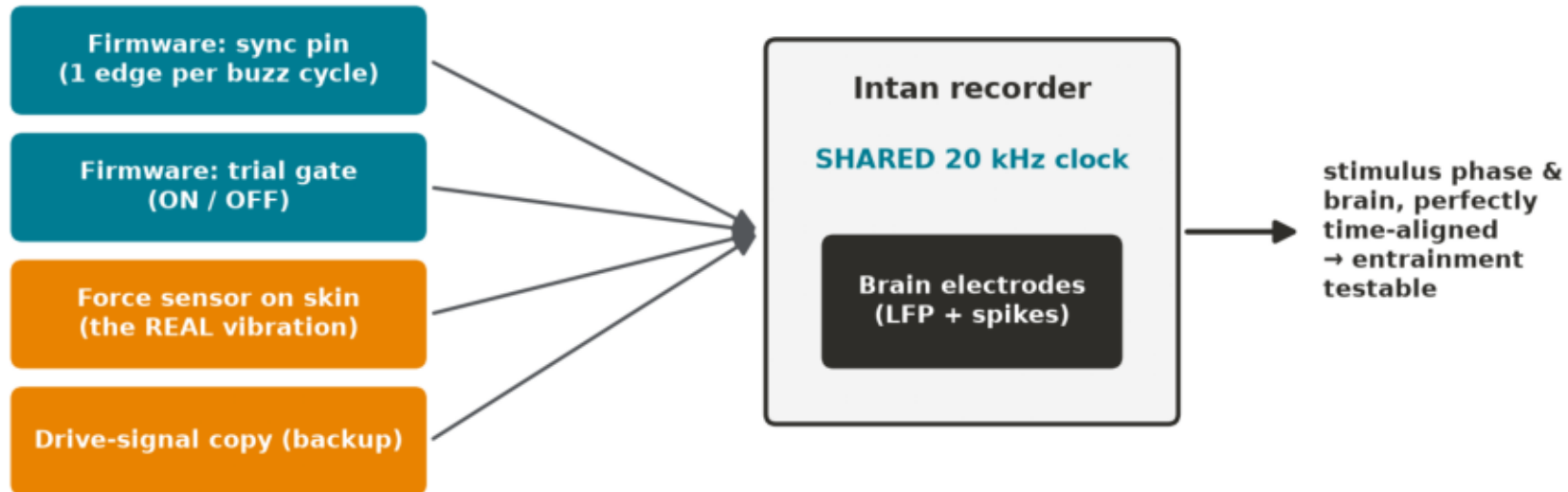
ise. Panel A is the BEST/densest window (cherry-picked to favor a carrier); Panel C is ALL data (typical). ch7 toggles $\sim 4\times/s$; a 26 Hz tactor toggles $26\times/s \rightarrow$ tactor does ~ 6 full



The recorded sync line updated at ~ 4 Hz, independent of carrier (identical pulse counts at 5 vs 26 Hz; ~ 78 expected at 26 Hz). No phase reference exists in the data.

Fix: redundant, clock-shared stimulus references

Fix: record the stimulus 3 ways on the acquisition clock



+ a 2-minute verification recording BEFORE the real session

Firmware per-cycle sync + a transduced force/accelerometer signal + a drive-signal copy, all on the acquisition clock — cross-checked, with pre-session verification.

Part 5 · How it fits the program

This study de-risks the program's core premise

It shows peripheral vibration reaches AD-relevant medial-temporal circuits and that those neurons are frequency-tuned (peak ~50 Hz) — directly answering the 'steady-state ≠ engagement' objection.

Key references

Lifestyle: Ngandu 2015 (FINGER) · Baker 2025 (US POINTER) · Livingston 2024 (Lancet Commission) · Barnes 2023 (MIND die — single-domain null). **Clearance:** Iliff 2012 · Xie 2013 · Mawuenyega 2010 · Murdock 2024 (40 Hz → glymphatic) · Aryal 2022 / Azadian 2025 (focused ultrasound). **Stimulation:** Tsai 2016 (GENUS) · Suk 2023 (40 Hz tactile) · Jiang-Xie-Kipnis 2024 (neural dynamics → CSF). **Guardrail:** Buzsáki / Soula (steady-state $\neq$ entrainment).

Thank you — questions

Happy to go into any figure, the artifact controls, the parameter plan, or the next-round instrumentation.